

Straight lines - Set 2 - Solutions

Specification MM3.1, M4.9, M4.10.

Answer key

Question	Answer	Topic
1	C	Straight lines
2	B	Straight lines
3	H	Straight lines

Worked solutions

Question 1 LIN-12 · Straight lines · spec MM3.1

The lines $y = mx + 4$ and $y = 3x - 2$ intersect at a point with positive x - and y -coordinates. Find the complete set of values of m .

Answer: C

$mx + 4 = 3x - 2 \Rightarrow x = \frac{6}{3-m}$, which is positive iff $m < 3$. Then $y = 3x - 2 > 0$ iff $x > \frac{2}{3}$, i.e. $\frac{6}{3-m} > \frac{2}{3}$, with $3 - m > 0$ this is $18 > 6 - 2m$, i.e. $m > -6$. So $-6 < m < 3$.

Full method - how and why

Two conditions on the intersection point. Find the intersection in terms of m , then impose positivity on each coordinate.

1 Find the intersection: $mx + 4 = 3x - 2 \Rightarrow x(m - 3) = -6 \Rightarrow x = \frac{6}{3 - m}$.
why: Keep the denominator as $3 - m$ so that positivity is easy to read.

2 $x > 0$ requires $3 - m > 0$, i.e. $m < 3$.
why: The numerator 6 is positive, so the sign of x is the sign of $3 - m$.

3 $y > 0$: since $y = 3x - 2$, this needs $x > \frac{2}{3}$, i.e. $\frac{6}{3 - m} > \frac{2}{3}$.
why: Use whichever line is simpler to express y in terms of x .

4 With $3 - m > 0$ we may multiply safely: $18 > 2(3 - m) \Rightarrow 18 > 6 - 2m \Rightarrow m > -6$.
why: Multiplying by a positive quantity is safe - and we already know $3 - m > 0$ from the first condition.

5 Combine: $-6 < m < 3$.
why: Both conditions must hold at once.

Where it goes wrong

- Checking only $x > 0$ and answering $m < 3$.
- Multiplying by $3 - m$ without noting its sign - legitimate here only because the first condition established it.

Takeaway Impose one condition at a time, and use the first to justify the algebra needed for the second.

Question 2 LIN-13 · Straight lines · spec MM3.1

The distance between the parallel lines $3x + 4y = 5$ and $3x + 4y = 20$ is

Answer: B

The lines are $3x + 4y = c$; the perpendicular direction is $(3, 4)$ of length 5. Distance $= \frac{|20 - 5|}{5} = 3$. (Alternatively, the point $(0, \frac{5}{4})$ on the first line is 3 from the second.)

Full method - how and why

Distance between parallel lines. Either use the coefficient formula or take any point on one line and measure to the other.

1 Note both lines have the same left-hand side $3x + 4y$, so they are parallel with normal direction $(3, 4)$, of length 5.

why: The vector of coefficients is perpendicular to the line; its length normalises the distance formula.

2 Use distance $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}} = \frac{|20 - 5|}{5}$.

why: This is the point-to-line formula applied to any point on the first line.

3 Evaluate: $\frac{15}{5} = 3$.

why: Check with a concrete point: $(0, \frac{5}{4})$ is on the first line, and its distance to $3x + 4y = 20$ is $\frac{|5 - 20|}{5} = 3$ ✓.

Where it goes wrong

- Answering 15 by forgetting to divide by $\sqrt{a^2 + b^2}$.
- Using $\sqrt{3^2 + 4^2} = 7$ (adding instead of squaring), giving $\frac{15}{7}$ - an offered option.

Takeaway For $ax + by = c_1$ and $ax + by = c_2$: distance $= \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$; the coefficients must match exactly first.

Question 3 LIN-10 · Straight lines · spec MM3.1

The line through $(a, 0)$ and $(0, b)$, with $a, b > 0$, passes through $(2, 3)$. Which of the following statements is/are true?

- I** $\frac{2}{a} + \frac{3}{b} = 1$
II $a > 2$ and $b > 3$
III The smallest possible value of ab is 24

Answer: H

The line is $\frac{x}{a} + \frac{y}{b} = 1$, so I holds. Since $\frac{2}{a} < 1$ and $\frac{3}{b} < 1$, II holds. $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$; setting $a = 2 + t$ gives $\frac{3(2+t)^2}{t} = 3(\frac{4}{t} + 4 + t) \geq 3(4 + 4) = 24$ by AM-GM (equality at $t = 2$, $a = 4$, $b = 6$). III holds.

Full method - how and why

The intercept form of a line, plus an optimisation. Statements I and II are quick; III needs a minimisation argument.

- I:** a line with intercepts a and b has equation $\frac{x}{a} + \frac{y}{b} = 1$; substituting $(2, 3)$ gives $\frac{2}{a} + \frac{3}{b} = 1$. **True.**
why: The intercept form is worth knowing precisely for questions like this.
- II:** since $\frac{2}{a}$ and $\frac{3}{b}$ are both positive and sum to 1, each is less than 1, so $a > 2$ and $b > 3$. **True.**
why: Two positive numbers summing to 1 are each below 1 - a clean argument with no algebra.
- III:** from I, $b = \frac{3a}{a-2}$, so $ab = \frac{3a^2}{a-2}$. Put $a = 2 + t$ with $t > 0$: $ab = \frac{3(2+t)^2}{t} = 3\left(\frac{4}{t} + 4 + t\right)$.
why: The substitution $a = 2 + t$ clears the awkward denominator and exposes the structure $\frac{4}{t} + t$.
- By AM-GM, $\frac{4}{t} + t \geq 2\sqrt{4} = 4$, with equality when $t = 2$. So $ab \geq 3(4 + 4) = 24$, attained at $a = 4$, $b = 6$. **True.**
why: $u + \frac{k}{u} \geq 2\sqrt{k}$ for positive u - here it gives both the bound and the case of equality, which is what 'smallest possible value' demands.
- All three are true.
why: Match to the options.

Where it goes wrong

- Assuming II fails because a and b 'could be anything' - the positivity constraint forces both bounds.
- Claiming III without checking equality is attainable; a lower bound is only the minimum if some case achieves it.

Takeaway $\frac{x}{a} + \frac{y}{b} = 1$ is the intercept form; and $t + \frac{4}{t} \geq 4$ (AM-GM) is the standard tool for such minimisations.